Sai Bhavadeesh Yarlagadda

Bangalore, Karnataka, India
8106263461 saibhavadeeshyarlagadda@gmail.com Sai Bhavadeesh Yarlagadda GitHub

Summary

AI Software Engineer with 4+ years of experience building scalable agentic AI, RAG, search, and MLOps solutions. Drive end-to-end development of strategic AI initiatives, including multi-agent systems, AI-powered search experiences, intelligent automation platforms, and production ML workflows. Skilled in translating business requirements into high-impact AI products, optimizing systems for cost and performance, and delivering solutions that enhance user & customer engagement, search quality, operational efficiency, and business growth.

Experience

Netcore Unbxd **September 2025 - Present**

AI Software Engineer — Agentic AI, AI, Python, Go, FastAPI, MLOps, Docker, Kubernetes *Bangalore, Karnataka*

- Owned the end-to-end development of an AI-powered Shopping Assistant, enabling conversational product discovery and information retrieval through Retrieval-Augmented Generation (RAG) and multi-agent workflows.
- Enhanced the Shopping Assistant with advanced capabilities including Visual Search, Product Comparison, Product Summarization, Search Insights, Document Understanding, and AI-powered Planning Tables, improving user shopping experiences.
- Designed and implemented configurable AI guardrails and site-specific response controls, enabling secure and flexible deployments across multiple e-commerce storefronts.
- Built Drishti, an internal relevance management platform using Django, Celery, and Redis, automating search relevance setup and evaluation workflows that were previously manual and time-consuming.
- Reduced AI-powered search relevance evaluation costs by 95% through token optimization, efficient model selection, and Gemini Batch Processing, while significantly improving processing throughput.
- Maintained and enhanced production ML pipelines for recommendation and classification models, including Top Sellers and Query Category Prediction, while introducing configurable store-level model training.
- Automated the Complete the Look (CTL) AI workflow by migrating it from a manual process to a scalable API-driven ML pipeline, improving operational efficiency and deployment reliability.
- Spearheaded strategic AI initiatives, including the Nike Customized Sneakers experience and an agent-driven AI Mode for search, contributing to successful customer deployments, industry showcases, and improvements in search relevance, conversion rates, and Average Order Value (AOV).

Tata Consultancy Services **July 2022 - September 2025**

Software Engineer — Python, MLOps, Flask, Docker, Kubernetes, Spring Boot *Bangalore, Karnataka*

- Designed and deployed end-to-end ML pipelines covering data ingestion, validation, transformation, training, and evaluation using Python and Apache Airflow.
- Developed scalable MLOps solutions with Flask, PostgreSQL, DVC, Docker, Kubernetes, and GitLab CI/CD to automate model training and deployment workflows.
- Built adaptive inference systems leveraging Multi-Armed Bandit (LinUCB) algorithms for dynamic model selection and optimized prediction performance.
- Engineered and deployed a heterogeneous network simulation platform integrating Mininet, Mininet-WiFi, and NS-3 (4G/5G) simulators on Kubernetes.
- Contributed to enterprise applications built with Spring Boot and Angular while collaborating in Agile teams to deliver secure and maintainable software solutions.

Education

Indian Institute of Information Technology, Kottayam **August 2018 – April 2022**

Bachelor of Technology in Computer Science, CGPA - 9.01 *Kottayam, Kerala*

Projects

Third Eye: Object Recognition and Speech Generation for Visually Impaired |

Python, PyTorch, OpenCV, PYTTSX3, gTTS **August 2021 - April 2022**

- Built a real-time assistive vision system using a custom-trained YOLOv5 model and speech synthesis to help visually impaired users interpret their surroundings.
- Co-authored and presented a research paper at the International Conference on Machine Learning and Data Engineering.

Technical Skills

Programming Languages: Python, Go, Java, TypeScript, Dart, HTML, CSS
AI/LLM Technologies: Agentic AI, RAG, Multi-Agent Systems, OpenAI Agents SDK, Google GenAI, Prompt Engineering
Backend & Frameworks: FastAPI, Django, Celery, Spring Boot
Cloud & Databases: AWS, Google Cloud Platform (GCP), PostgreSQL, MongoDB, Redis
DevOps & MLOps: Docker, Kubernetes, Helm, Kubeflow, GitHub Actions, CI/CD
Architecture & Practices: Microservices, Distributed Systems, MLOps, Agile/Scrum
Tools: Cursor AI, Git, JIRA
Additional Technologies: Flutter, React Native